

Exercise 2.2

The function

$$G(x) = \frac{18000 - 110x - x^2}{18000}$$

has been proposed as the survival function $S_0(x)$ for a mortality model.

- (a) What is the implied limiting age ω ?
- (b) Verify that the function G satisfies the criteria for a survival function.
- (c) Calculate ${}_{20}p_0$.
- (d) Determine the survival function for a life aged 20.
- (e) Calculate the probability that a life aged 20 will die between ages 30 and 40.
- (f) Calculate the force of mortality at age 50.

Solution

(a)

The limiting age ω is the age at which the survival function becomes zero. Thus, we solve

$$18000 - 110x - x^2 = 0.$$

Equivalently,

$$x^2 + 110x - 18000 = 0.$$

Using the quadratic formula,

$$\begin{aligned} x &= \frac{-110 \pm \sqrt{110^2 - 4(1)(-18000)}}{2} \\ &= \frac{-110 \pm \sqrt{84100}}{2} \\ &= \frac{-110 \pm 290}{2}. \end{aligned}$$

Therefore,

$$x_1 = 90, \quad x_2 = -200.$$

Since age must be nonnegative, the implied limiting age is

$$\boxed{\omega = 90}.$$

(b)

We verify the criteria for a survival function on the domain $[0, 90]$. First,

$$G(0) = \frac{18000 - 110(0) - 0^2}{18000} = 1.$$

Also,

$$G(90) = \frac{18000 - 110(90) - 90^2}{18000} = 0.$$

Next, for $0 \leq x \leq 90$,

$$G'(x) = \frac{-110 - 2x}{18000} < 0.$$

Thus G is nonincreasing on $[0, 90]$.

Finally, because $G(0) = 1$, $G(90) = 0$, and G is decreasing on $[0, 90]$, we have

$$0 \leq G(x) \leq 1 \quad \text{for } 0 \leq x \leq 90.$$

Therefore, G satisfies the criteria for a survival function on the domain

$$\boxed{[0, 90]}.$$

(c)

The probability that a newborn survives to age 20 is

$${}_{20}p_0 = \frac{S_0(20)}{S_0(0)}.$$

Since $S_0(x) = G(x)$ and $S_0(0) = 1$,

$$\begin{aligned} {}_{20}p_0 &= G(20) \\ &= \frac{18000 - 110(20) - 20^2}{18000} \\ &= \frac{18000 - 2200 - 400}{18000} \\ &= \frac{15400}{18000} \\ &= \frac{77}{90}. \end{aligned}$$

Thus,

$$\boxed{{}_{20}p_0 = \frac{77}{90} \approx 0.855556}.$$

(d)

For a life aged 20, the survival function gives the probability of surviving a further t years:

$${}_tp_{20} = \frac{S_0(20+t)}{S_0(20)}, \quad 0 \leq t \leq 70.$$

Using $S_0(x) = G(x)$ and $S_0(20) = 77/90$, we get

$$\begin{aligned} {}_tp_{20} &= \frac{\frac{18000 - 110(20+t) - (20+t)^2}{18000}}{\frac{77}{90}} \\ &= \frac{18000 - 110(20+t) - (20+t)^2}{15400} \\ &= \frac{15400 - 150t - t^2}{15400}. \end{aligned}$$

Therefore, the survival function for a life aged 20 is

$${}_tp_{20} = 1 - \frac{3t}{308} - \frac{t^2}{15400}, \quad 0 \leq t \leq 70.$$

(e)

For a life aged 20, dying between ages 30 and 40 means surviving from age 20 to age 30 and then dying before age 40. Hence,

$${}_{10|10}q_{20} = {}_{10}p_{20} {}_{10}q_{30}.$$

First, using the survival function from part (d),

$$\begin{aligned} {}_{10}p_{20} &= 1 - \frac{3(10)}{308} - \frac{10^2}{15400} \\ &= \frac{69}{77}. \end{aligned}$$

Next,

$${}_{10}p_{30} = \frac{S_0(40)}{S_0(30)}.$$

Now,

$$\begin{aligned} S_0(30) &= \frac{18000 - 110(30) - 30^2}{18000} = \frac{23}{30}, \\ S_0(40) &= \frac{18000 - 110(40) - 40^2}{18000} = \frac{2}{3}. \end{aligned}$$

Therefore,

$$\begin{aligned} {}_{10}p_{30} &= \frac{2/3}{23/30} = \frac{20}{23}, \\ {}_{10}q_{30} &= 1 - {}_{10}p_{30} = 1 - \frac{20}{23} = \frac{3}{23}. \end{aligned}$$

Thus,

$$\begin{aligned} {}_{10|10}q_{20} &= {}_{10}p_{20} {}_{10}q_{30} \\ &= \frac{69}{77} \cdot \frac{3}{23} \\ &= \frac{9}{77}. \end{aligned}$$

So the required probability is

$$\boxed{{}_{10|10}q_{20} = \frac{9}{77} \approx 0.116883}.$$

(f)

The force of mortality at age x is

$$\mu_x = -\frac{1}{S_0(x)} \frac{d}{dx} S_0(x).$$

Since $S_0(x) = G(x)$,

$$\mu_{50} = -\frac{G'(50)}{G(50)}.$$

Now,

$$G'(x) = \frac{-110 - 2x}{18000},$$

so

$$G'(50) = \frac{-110 - 2(50)}{18000} = \frac{-210}{18000}.$$

Also,

$$G(50) = \frac{18000 - 110(50) - 50^2}{18000} = \frac{10000}{18000}.$$

Therefore,

$$\begin{aligned} \mu_{50} &= -\frac{-210/18000}{10000/18000} \\ &= \frac{210}{10000} \\ &= 0.021. \end{aligned}$$

Thus,

$$\boxed{\mu_{50} = 0.021}.$$